

Vector Fields and Line Integrals

ANSWER KEY



Vector fields and divergence

- 1 For the vector field $\mathbf{F}(x, y) = \langle x^2y, xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(xy^2) = 2xy + 2xy = 4xy.$$

- 2 For $\mathbf{F}(x, y, z) = \langle y, -x, z^2 \rangle$, find the curl $\nabla \times \mathbf{F}$.

$$\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(-x), \frac{\partial}{\partial z}(y) - \frac{\partial}{\partial x}(z^2), \frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y) \right\rangle = \langle 0, 0, -2 \rangle.$$

Line integrals

- 3 Evaluate the line integral $\int_C (x + y) ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.

Parametrize $x = 3t$, $y = 4t$, $0 \leq t \leq 1$: $ds = 5 dt$. $\int_0^1 (3t + 4t)(5) dt = 35 \int_0^1 t dt = \frac{35}{2}$.

- 4 Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \leq t \leq 1$.

$\mathbf{F}(\mathbf{r}(t)) = \langle t^2, t \rangle$, $\mathbf{r}'(t) = \langle 1, 2t \rangle$: $\int_0^1 (t^2 + 2t^2) dt = \int_0^1 3t^2 dt = 1$.